Factors affecting solubility

Solubility tells us how much of something can dissolve in something else. There is a limit to how much **solute** will dissolve in a **solvent**. When no more solute will dissolve, we say that the solution has become **saturated**.

☞ **1** Use the words in the word bank to complete the following statements.

Solubility is affected by the following factors:

- whether or not the solution is _____, and how quickly
- the type of _____ the solute is dissolved in
- the _____ of the solute particles. Smaller-sized particles dissolve more _____
- the _____ of the solvent.

Word bank
size
solvent
quickly
temperature
stirred

Different substances have different solubilities, and solutes become more soluble at higher temperatures.

Temperature	Potassium chloride Solubility g/100g water	Copper sulfate Solubility g/100g water	Sodium chloride Solubility g/100g water
10°C	31	19	34.5
30°C	37	24	36
50°C	42	33	37
70°C	49	48	38
90°C	53	65	39.5

2 Which substance has the highest solubility at 50°C? _____

3 Which substance's solubility is the least affected by changes in temperature? _____

4 What is the solubility of potassium chloride at 30°C? _____

5 Which substance has a solubility of 48g/100g water at 70°C? _____

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